

Q: 2 (i) (or)
A: 2 (i) (or)

① $-8^{\frac{1}{3}}$

$(-2^3)^{\frac{1}{3}}$

$= \frac{1}{2} \quad \text{or} \quad -2$

② ~~$(\sqrt{25})^4$
 $((25)^{\frac{1}{2}})^{4 \times 2}$
 $(25)^2$
 $(5^2)^2$
 5^4~~

$(\sqrt{25})^4$
 $(5)^4$
625

③ $(x^{-3})^2 \cdot (x^0)^5$
 $x^{-6} \cdot 1$
 $x^{-6} \quad \text{or} \quad \frac{1}{x^6}$

Q: 2 (ii)
A: 2 (ii)

- ① True
- ② True
- ③ False

Q: 2 (iii) (or) _____
A: 2 (iii) (or) _____

① 7.05×10^{-4}
 $= 0.000705$

② 2.31×10^6
 $2,310,000$

③ 0.00000000512
 5.12×10^{-8}

Q: 2 (iv) _____
A: 2 (iv) _____

① $\log_n 125 = 3$

$125 = n^3$

$(5)^3 = n^3$

$5 = n$

② $\log_4 n = -3/2$

$n = 4^{-3/2}$

$n = ((2)^2)^{-3/2}$

$n = 2^{-3}$

$n = 1/8$

$n = \frac{1}{8}$

$$\textcircled{b} \log_x (x^2 - 3) = \log_x 13$$

$$x^2 - 3 = 13$$

$$x^2 = 13 + 3$$

$$x^2 = 16$$

$$x^2 = 4^2$$

$$x = 4$$

$$\text{Q: } 2(\checkmark) \quad \text{A: } 2(\checkmark)$$

$$\textcircled{1} 4580$$

$$\log 4580$$

$$4.580 \times 10^3$$

$$\text{characteristics: } -3$$

$$\text{mantissa: } 6609$$

$$= 3.6609$$

$$\textcircled{2} 0.00093$$

$$9.3 \times 10^{-4}$$

$$\text{characteristic: } -\bar{4}$$

$$\text{mantissa: } 9685$$

$$= \bar{4}.9685$$

$$\textcircled{3} 54$$

$$5.4 \times 10^1$$

$$\text{characteristic: } 1$$

$$\text{mantissa: } 7324$$

$$1.7324$$

Q.2 (vi) _____
A: 2 (vi) _____

i) $x^4 + 4m^4$

$$(x^2)^2 + (2m^2)^2$$

$$= (x^2)^2 + 2(x^2)(2m^2) + (2m^2)^2$$

$$(x^2 + 2m^2)^2 - (2mx)^2$$

$$(x^2 + 2m^2 - 2mx)(x^2 + 2m^2 + 2mx)$$

~~Q.2~~

ii) $3x^2 + 11x + 6$

$$3x^2 + 9x + 2x + 6$$

$$3x(x+3) + 2(x+3)$$

$$(3x+2)(x+3)$$

Q.2 (vii) _____
A: 2 (vii) _____

① $x^2 - 49$

$$(x)^2 - (7)^2$$

$$(x-7)(x+7)$$

②

$$x^2 - 4x + 21$$

$$x^2 + 3x - 7x + 21$$

$$x(x-3) - 7(x-3)$$

$$(x-3)(x-7)$$

$$\therefore \text{HCF} = (x-7)$$

Q: 2(viii)

A: 2(vii)

$$\frac{(2n+3)}{5} = \frac{(3-4n)}{8}$$

~~$$\frac{2n+3}{5} = \frac{3-4n}{8}$$

$$\frac{2n+3}{5} \times 8 = \frac{3-4n}{8} \times 5$$

$$\frac{16n+24}{40} = \frac{15-20n}{40}$$~~

$$\frac{2n+3}{5} = \frac{3-4n}{8}$$

$$5(3-4n) = 8(2n+3)$$

$$15 - 20n = 16n + 24$$

$$15 - 24 = 16n + 20n$$

$$9 = 36n$$

$$36n = 9$$

$$n = \frac{9}{36}$$

$$n = \frac{1}{4}$$

$$n = \frac{1}{4}$$

$$4$$

Q: 2(ix)

A: 2 (ix)

$$|4n-3| = 13$$

$$4n-3 = 13$$

$$4n = 13+3$$

$$4n = 16$$

$$n = \frac{16}{4}$$

$$n = 4$$

$$n = 4$$

$$\{4, -\frac{5}{2}\}$$

$$4n-3 = -13$$

$$4n = 3-13$$

$$4n = -10$$

$$n = \frac{-10}{4}$$

$$n = -\frac{5}{2}$$

$$n = -\frac{5}{2}$$

Q: 2(n)
A: 2(n)

$A(1, -2) \quad B(7, 8) \quad C(4, 3)$

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(7 - 1)^2 + (8 - (-2))^2}$$

$$= \sqrt{6^2 + (8 + 2)^2}$$

$$= \sqrt{36 + (10)^2}$$

$$= \sqrt{36 + 100}$$

$$= \sqrt{136}$$

$$= 2\sqrt{34}$$

$$d_{BC} = \sqrt{(4 - 7)^2 + (3 - 8)^2}$$

$$= \sqrt{(-3)^2 + (-5)^2}$$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$d_{CA} = \sqrt{(4 - 1)^2 + (3 - (-2))^2}$$

$$= \sqrt{3^2 + (3 + 2)^2}$$

$$= \sqrt{9 + (5)^2}$$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$\frac{BC + AC}{2\sqrt{34}} = \frac{AB}{2\sqrt{34}}$$

$$\frac{\sqrt{34} + \sqrt{34}}{2\sqrt{34}} = \frac{2\sqrt{34}}{2\sqrt{34}}$$

Collinear.

$$\begin{aligned} Q &: 2(x_1) \\ A &: 2(x_1) \end{aligned}$$

Midpoint P of AB :-

$$\left(\frac{-1+7}{2}, \frac{7+(-3)}{2} \right)$$

$$\frac{6}{2}, \frac{7-3}{2}$$

$$\frac{6}{2}, \frac{4}{2}$$

$$\frac{3}{K=2}, 2$$

P(3, 2) distance PA :-

$$\begin{aligned} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(3 - (-1))^2 + (2 - 7)^2} \\ &= \sqrt{(3+1)^2 + (-5)^2} \\ &= \sqrt{(4)^2 + (-5)^2} \\ &= \sqrt{16 + 25} \end{aligned}$$

$\sqrt{41}$ units.

∴ Long Question:-

Q: 3(i)(or)

A: 3(i)(or)

$$\log n = \frac{\sqrt[3]{8.59} \times (55.6)^2}{2.51 \times 12.12}$$

$$\log n = \frac{(8.59)^{\frac{1}{3}} \times (55.6)^2}{2.51 \times (2.12)^{\frac{1}{2}}}$$

$$\log 8.59^{\frac{1}{3}} + \log 55.6^2 - \log 2.51$$

$\bullet \log 2.12^{\frac{1}{2}}$

$$\frac{1}{3} \log 8.59 + 2 \log 55.6 - \log 2.51$$

$\bullet - \frac{1}{2} \log 2.12$

$$\frac{1}{3} (0.9339) + 2(1.7450) - 0.3996$$

$- \frac{1}{2} (0.3263)$

$$0.3113 + 3.49 - 0.3996 - 0.16315$$

$$3.8013 - 0.23585$$

$$3.56545$$

$$\text{antilog} \approx 1732$$

Q: 3 (ii) (or)
A: 3 (ii) (or)

(a) $4a^2 + 12ab + 9b^2$

$$(2a)^2 + 2(2a)(3b) + (3b)^2$$

$$(2a + 3b)^2$$

(b) Let $x^2 + 3x = a$

$$(a-4)(a+5) + 8$$

$$a(a+5) - 4(a+5) + 8$$

$$a^2 + 5a - 4a - 20 + 8$$

$$= a^2 + a - 12$$

$$= a^2 - 3a + 4a - 12$$

$$= a(a-3) + 4(a-3)$$

$$(a+4)(a-3)$$

Putting $x^2 + 3x$ back :-

$$(x^2 + 3x + 4)(x^2 + 3x - 3)$$

(c) $8a^3 - 125b^3 - 2a + 5b$

$$(2a)^3 - (5b)^3 - 2a + 5b$$

$$(2a - 5b)((2a)^2 + (2a)(5b) + (5b)^2) - 2a + 5b$$

$$(2a - 5b)(4a^2 + 10ab + 25b^2) - 2a + 5b$$

$$(2a - 5b)(4a^2 + 10ab + 25b^2 - 1)$$

Q.3(iii)

A.3(iii)

$$\frac{y+1}{3} + \frac{y+1}{2} = 2 - \frac{y+3}{2}$$

$$\frac{(y+1)^{\times 2}}{3^{\times 2}} + \frac{(y+1)^{\times 3}}{2^{\times 3}} = \frac{2^{\times 2}}{1^{\times 2}} - \frac{y+3^{\times 1}}{2^{\times 1}}$$

$$\frac{2y+2+3y+3}{6} = \frac{4-y+3}{2}$$

$$\frac{2y+3y+3+2}{6} = \frac{4+3-y}{2}$$

$$\frac{5y+5}{6} = \frac{7-y}{2}$$

$$6(7-y) = 2(5y+5)$$

$$42-6y = 10y+10$$

$$42-10 = 10y+6y$$

$$32 = 16y$$

$$16y = 32$$

$$y = \frac{32}{16}$$

$$y = 2$$

$$y = 2$$

Q.3(iv)

A:3(iv)

$$\begin{aligned}d_{AB} &= \sqrt{(6-0)^2 + (0-0)^2} \\&= \sqrt{(6)^2 + (0)^2} \\&= \sqrt{36+0} \\&= \sqrt{36} \\&= 6\end{aligned}$$

$$\begin{aligned}d_{BC} &= \sqrt{(6-6)^2 + (4-0)^2} \\&= \sqrt{(0)^2 + (4)^2} \\&= \sqrt{0+16} \\&= \sqrt{16} \\&= 4\end{aligned}$$

$$\begin{aligned}d_{CD} &= \sqrt{(0-6)^2 + (4-4)^2} \\&= \sqrt{(-6)^2 + (0)^2} \\&= \sqrt{36+0} \\&= \sqrt{36} \\&= 6\end{aligned}$$

$$\begin{aligned}d_{AD} &= \sqrt{(0-0)^2 + (4-0)^2} \\&= \sqrt{(0)^2 + (4)^2} \\&= \sqrt{0+16} \\&= \sqrt{16} \\&= 4\end{aligned}$$

opposite side

$$AB = CD = 6$$

$$BC = AD = 4$$

② Diagonals:-

$$\begin{aligned} AC &= \sqrt{(6-0)^2 + (4-0)^2} \\ &= \sqrt{(6)^2 + (4)^2} \\ &= \sqrt{36 + 16} \\ &= \sqrt{52} \end{aligned}$$

$$\begin{aligned} BD &= \sqrt{(0-6)^2 + (4-0)^2} \\ &= \sqrt{(-6)^2 + (4)^2} \\ &= \sqrt{36 + 16} \\ &= \sqrt{52} \end{aligned}$$

midpoint of AC :-

$$\frac{0+6}{2}, \frac{0+4}{2}$$

$$\frac{6}{2}, \frac{4}{2}$$

$$3, 2$$

midpoint of BC :-

$$\frac{6+0}{2}, \frac{0+4}{2}$$

$$\frac{6}{2}, \frac{4}{2}$$

3 9 2

ABCD is a rectangle.

MCQs:-

1. B
2. C
3. B
4. C
5. B
6. A
7. C
8. B
9. A
10. C
11. B
12. B